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Xiaoqing: A Q&A model for glaucoma based on LLMs

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ABSTRACT

Glaucoma is one of the leading cause of blindness worldwide. Individuals affected by glaucoma, including patients and their family members, frequently encounter a deficit in dependable support beyond the confines of clinical environments. Seeking advice via the internet can be a difficult task due to the vast amount of disorganized and unstructured material available on these sites, nevertheless. This research explores how Large Language Models (LLMs) can be leveraged to better serve medical research and benefit glaucoma patients. We introduce Xiaoqing, a Natural Language Processing (NLP) model specifically tailored for the glaucoma field, detailing its development and deployment. To evaluate its effectiveness, we conducted two forms of experiments: comparative and experiential. In the comparative analysis, we presented 22 glaucoma-related questions in simplified Chinese to three medical NLP models (Xiaoqing LLMs, HuaTuo, Ivy GPT) and two general models (ChatGPT-3.5 and ChatGPT-4), covering a range of topics from basic glaucoma knowledge to treatment, surgery, research, management standards, and patient lifestyle. Responses were assessed for informativeness and readability. The experiential experiment involved glaucoma patients and non-patients interacting with Xiaoqing, collecting and analyzing their questions and feedback on the same criteria. The findings demonstrated that Xiaoqing notably outperformed the other models in terms of informativeness and readability, suggesting that Xiaoqing is a significant advancement in the management and treatment of glaucoma in China. We also provide a Web-based version of Xiaoqing, allowing readers to directly experience its functionality. The Web-based Xiaoqing is available at https://qa.glaucoma-assistant.com//qa.

1. Introduction

Glaucoma, an ocular condition characterized by elevated intraocular pressure leading to optic nerve damage and subsequent visual field impairment, is a significant global health concern. The 2019 World Vision Report by the World Health Organization estimated that approximately 76 million individuals aged 40 to 80 years worldwide would be affected by glaucoma by the year 2020 [1]. The number of glaucoma patients in China exceeds 21 million in 2020 [2], positioning it as the country with the highest prevalence of this condition globally. Vision rehabilitation is very effective in improving functioning for people with an irreversible vision loss that can be caused by eye conditions such as diabetic retinopathy [3]. Estimates indicate that in China, approximately 9.4 million people aged 40 and above are afflicted with glaucomatous optic neuropathy. Within this demographic, about 5.2 million individuals, constituting 55%, suffer from blindness in at least one eye, while nearly 1.7 million, representing 18.1%, experience bilateral blindness [4].

Consequently, glaucoma is recognized globally as the leading cause of irreversible blindness. Given that the onset of most types of glaucoma is exceptionally subtle and the disease's symptoms are often inconspicuous, there is a prevalent lack of disease awareness among many patients. In the case of primary open-angle glaucoma, the most

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prevalent form of the disease, it is estimated that nearly 50% of those affected are unaware of their condition [5]. The combination of low public awareness and the asymptomatic nature of glaucoma in its early stages contributes to delayed medical consultations and escalating financial burdens. The 2019 Global Burden of Disease study estimated that vision impairment, including conditions like glaucoma, resulted in an economic loss of $410.7 billion in 2020 [6]. These findings underscore the critical need for public education about glaucoma.

When individuals with glaucoma or their associates face challenges beyond the hospital, they frequently turn to the internet for solutions, often utilizing standard search engines for this purpose [7]. However, when addressing professional or intricate matters, conventional search engines present certain limitations. For instance, they often yield generalized information that may fall short in addressing the nuanced or detailed inquiries pertinent to specialized fields. In the context of medical domains such as glaucoma, the variability in the quality of information provided by these search engines necessitates users to invest effort in discerning and validating the information's accuracy and reliability. Additionally, these search engines exhibit limited interactivity, impeding users from articulating their specific needs or engaging in dialogic exchanges to acquire further information.

In the United States, medical service providers are proactively promoting awareness about glaucoma through their websites. These medical establishments have classified relevant information on glaucoma, which educates the public about the symptoms, different types, and causes of the disease, such as the Mayo Clinic, Centers for Disease Control and Prevention [8], or list it in a question-and-answer format, such as National Eye Institute [9], Cleveland Clinic [10], American Academy of Ophthalmology [11], etc. It is important to note that currently in China, there is a lack of a comprehensive platform for disseminating glaucoma knowledge to glaucoma patients. While efforts are being made to create accessible websites, the construction of such platforms is still in its early stages.

As such, there is an urgent need to establish an easy to access knowledge-based platform for glaucoma that is tailored to the Chinese population. By doing so, more people can gain access to important information about glaucoma, including its symptoms, causes, treatments and access the latest research and developments in the field [12], which can provide help to glaucoma patients and guide them in dealing with issues such as diet, exercise, treatment and prevention in daily life.

Professional domain-specific LLMs, being trained on datasets tailored to particular medical fields, are adept at delivering more accurate and pertinent information [13]. Their proficiency in interpreting and handling complex contextual data means they are better equipped to comprehend medical jargon and the specific requirements associated with it, thereby furnishing more precise insights in the realm of glaucoma. Moreover, these models have the capability to amalgamate knowledge from diverse sources, presenting a more holistic viewpoint. This contrasts with traditional search engines, which often necessitate users to manually synthesize information from various search outcomes. Notably, prior LLMs had not been explicitly focused on the glaucoma domain [14].

This study introduces the LLMs model specifically for glaucoma — Xiaoqing, shows Xiaoqing's model architecture and characteristics, and compares four models (including two general NLP models ChatGPT-3.5 and ChatGPT-4.0 [15], that are currently attracting much attention. Two professional medical NLP models HuaTuo [16] and Ivy GPT [17]. Subsequently, we elaborate on the performance and scenarios faced by these models in addressing issues related to glaucoma [18].

2. Related work

In recent years, the rapid development of computer algorithms, hardware and program architecture has improved the performance of deep learning and NLP models, making it easier to train large-scale neural networks [19]. The popularity of the Internet and the prevalence of social media have produced a large number of data, which in turn provide valuable data resources for training models. Therefore, in medical data processing and conversational models, AI has been regarded as a key tool for solving complex problems [20].

NLP enables computers to understand and process medical texts, helping doctors quickly understand the latest research and assist in disease diagnosis [21]; for patients, AI based on NLP technology can provide help outside the clinic, assisting patients. LLMs are AI models based on deep learning with large-scale parameters and large amounts of training data [22,23]. These models are able to learn the syntax, semantics and context of natural language, enabling them to understand and generate corresponding text. LLMs provide powerful tools for processing natural language. They have been widely used in NLP tasks and provide opportunities for improvement and innovation in various applications.

Xiaoqing is a NLP model for the field of glaucoma in simplify Chinese. Fig. 1 introduce the architecture of the entire model of Xiaoqing LLMs. It is improved based on the ChatGLM-6B model and fine-tuned by LoRA [24] and Freeze [25]. The dataset consists of relevant papers in the glaucoma field in recent years, and glaucoma field knowledge reviewed by specialist doctors [26,27]. The UI design part adopts the Vue framework and is embedded in the WeChat mini program. The following will introduce the various parts and deployment of the entire model [28].

2.1. ChatGLM-6B

In recent years, the popular open source LLMs model architectures that support Chinese are none other than Alpaca [29] and General Language Model (GLM) [30]. As a universal pre-training framework based on the Transformer architecture, GLM unifies the pre-training goals of different tasks into the Autoregressive Blank Infilling task, making the model perform better in both NLP understanding and text generation.

The purpose of this study is to build a focused and accurate model for the field of glaucoma, rather than for the general field, which requires a more refined and compact model. ChatGLM-6B, as a popular open source model in recent years, is a lightweight model based on the GLM architecture, supports bilingual dialogue in Chinese and English, and has 6.2 billion parameters. Parameters of this order of magnitude have no redundant parts and alleviate the development pressure as much as possible. They are very suitable for development and application in this limited field. Therefore, the model of this study was chosen to be fine-tuned on ChatGLM-6B.

2.2. Langchain

RAG (Retrieval-Augmented Generation) is frequently used in LLMs to improve the model's performance on a particular problem and to increase the model's capacity to handle information retrieval and generation tasks. Retrieval strategies are combined with the capacity to create models in order to introduce and make use of outside knowledge. The design of LLMs + LangChain is employed in the majority of knowledge bases that are now on the market, and the Langchain technique used in this study is based on the RAG technique. The process of vectorizing and matching user questions to text vectors in the database is illustrated in Fig. 2, the vector similarity algorithm is then used to identify the most similar text vectors, and the context corresponding to these vectors is inserted into the Prompt along with the user's questions. Finally, the questions are submitted to LLMs for the creation of the answers.

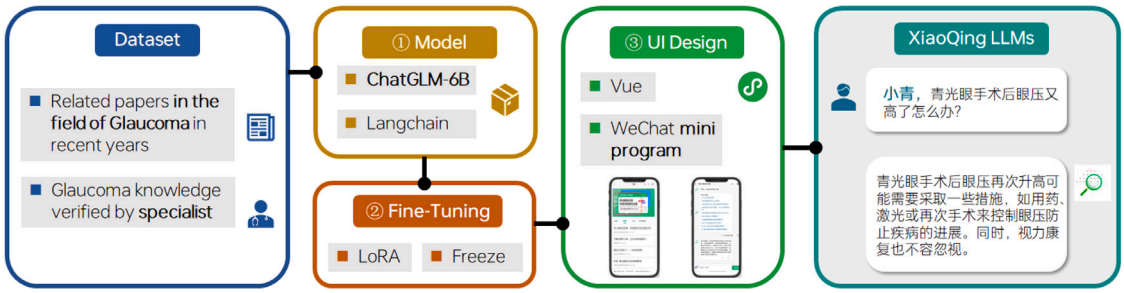

Fig. 1. Model architecture diagram of Xiaoqing. Xiaoqing's dataset is composed of recent papers in the field of glaucoma and knowledge information reviewed by glaucoma experts. The base model used is the lightweight ChatGLM-6B model, combined with Langchain to solve long sentence problems. It also employs LoRA and Freeze for fine-tuning parameters, and finally, the model is embedded into a WeChat mini-program using the Vue framework.

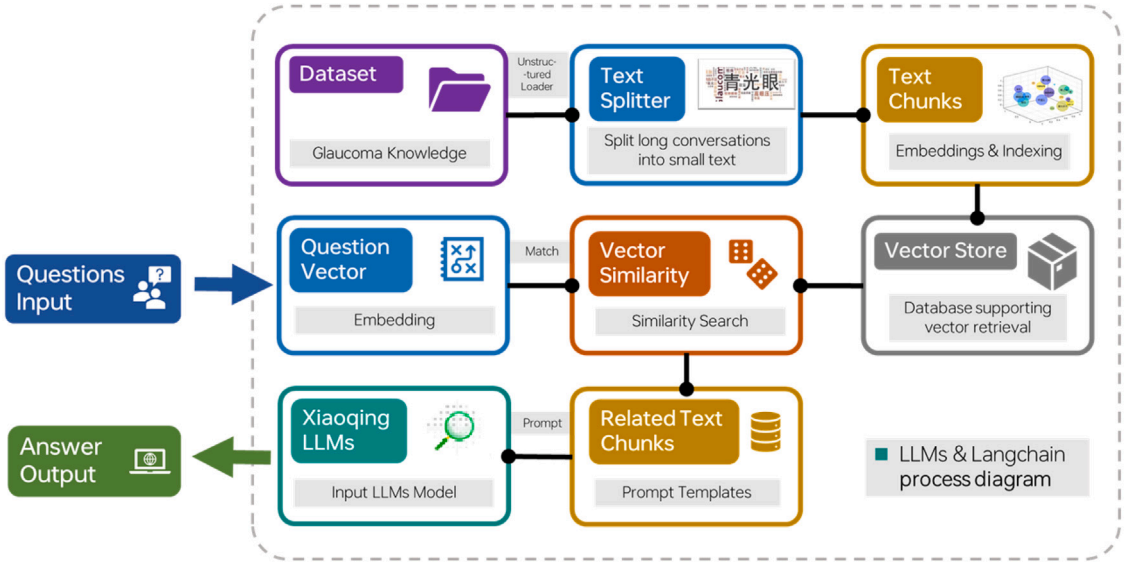

Fig. 2. Model data flow diagram of Xiaoqing. After a user asks a question, the question data is input into LLMs & Langchain. It is first decomposed into a questionvector, then undergoes vector similarity processing. This is followed by prompting after related text chunks, entering the LLM to generate an answer, and then outputting the response.

2.3. Fine-tuning

Choosing an LLMs model framework is only the first step in building Xiaoqing, the second step is to fine-tune the model. Since the LLMs model has many parameters, manually adjusting the parameters will consume a lot of manpower and material resources from beginning to end. Moreover, this study only focuses on the field of glaucoma, thus fine-tuning is introduced. There are currently three mainstream methods for fine-tuning large models: the Freeze, the P-Tuning [31] and the LoRA. Xiaoqing fine-tunes the low-rank adapter and the fully connected layer using LoRA and Freeze, respectively.

By using a low-rank matrix to control the model's weights, LoRA effectively fine-tunes the model while maintaining the majority of the pre-training weights with a minimal amount of parameter changes. Combined with Freeze, this technique further minimizes the computational load during training by requiring only a small set of parameters (the low-rank matrix) to be updated while the majority of the other parameters are frozen.

In contrast, P-Tuning realizes fine-tuning by adjusting the prompt, which relies on effective prompt design and can increase the complexity and arithmetic requirements of fine-tuning. So in this study, LoRA is chosen to combine fine-tuning model with Freeze. It is possible to optimize the results for problems in the glaucoma field while ensuring that Xiaoqing, also takes into account the basis of the original model as a medical LLMs.

The fine-tuned dataset uses the Alpace, the parameters for LoRA are r is 8, lora_alpha is 32, lora_dropoutis 0.1. freeze_module_name is "layers.27", "layers.26", "layers.25", "layers.24" in Freeze's parameters, train only the fully connected layer.

2.4. UI design

The objective of this study is to develop a LLM that is more accessible and convenient for glaucoma patients and associated groups. Given that the target demographic largely overlaps with middle-aged and elderly individuals, it is imperative to design an interactive UI that is simple and facilitates easy access to help. To meet these requirements, the study has opted to leverage the WeChat mini-program feature. WeChat, being the largest Chinese social platform, is utilized daily by tens of millions for its mini-program functionalities [32], making it a strategic choice for reaching the intended audience effectively. For end-users, this platform offers a highly efficient avenue for promoting the Xiaoqing extensively. It also fosters an environment conducive to user-centric dissemination and influence augmentation. The interactive user interface is meticulously crafted for simplicity, allowing for seamless operation through intuitive touchscreen interactions. On the developer front, the choice of small programs presents a strategic advantage. These programs are characterized by abbreviated development cycles and minimal learning requisites, rendering them particularly apt for deployment in niche, specialized domains. This synergy between user accessibility and developer convenience underscores the project's commitment to innovation in the field of language model applications.

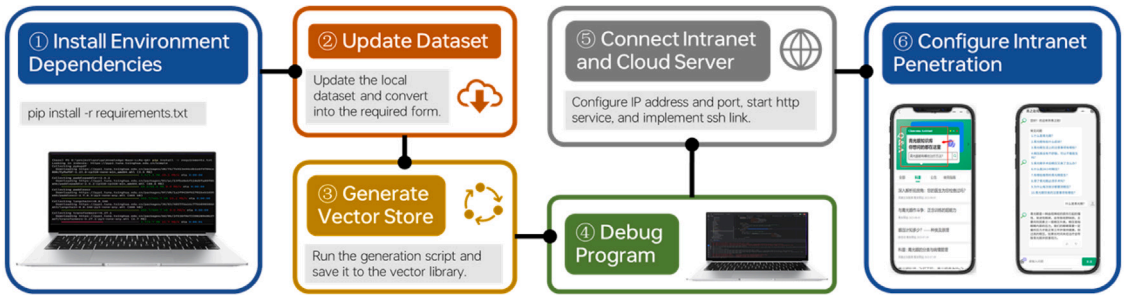

Fig. 3. Deployment flowchart of Xiaoqing on windows computers. The first step is to use pip to install the corresponding environment dependencies. The second step is to update the latest glaucoma dataset and use the generate vector script to generate a vector store. Then open up the internal and external transparent transmission, configure http, and Xiaoqing is deployed.

As the current mainstream front-end framework, Vue is a progressive JavaScript framework that provides developers with a complete set of solutions. The Vue framework is relatively lightweight. It is used in projects in the glaucoma field combined with WeChat applet, and the learning cost is relatively low. Using the Vue framework to build mini-programs can improve development efficiency and code quality, while expanding the functionality of the mini-program.

2.5. Deployment of xiaoqing

To facilitate development, Xiaoqing is built on a Windows 11 system. The computer hardware is equipped with an Nvidia RTX 4090 graphics card and 32G memory. As shown in the flow chart in Fig. 3, this section will introduce the deployment of Xiaoqing. The entire deployment is divided into six parts:

1. Install Environment Dependencies: First, we need to set up the compilation and runtime environment. It is recommended to use the requirements file for environment configuration.
2. Update Dataset: The dataset contains the latest consultations and papers in the field of glaucoma, as well as glaucoma knowledge curated by ophthalmic experts [33,34]. The practice of using professional data sources for a specialized domain ensures that the data is less susceptible to contamination from the source, thereby reducing the probability of generating errors in responses. Additionally, as a medical-oriented LLMs focusing on glaucoma, Xiaoqing's dataset is more comprehensive and detailed than public datasets, with regular updates ensuring that the data remains current. This ensures that Xiaoqing's user experience is better than other models.
3. Generate Vector Store: The primary purpose of generating vector stores is to represent text or language as numerical vectors, thereby capturing semantic information and language relationships. This enables LLMs to better understand and process natural language, showcasing the powerful performance across various NLP tasks.
4. Connect Intranet and Cloud Server: Xiaoqing employs Fast Reverse Proxy (FRP) for intranet penetration. In this setup, the locally hosted application serves as the client, while a cloud server acts as the server. By utilizing the FRP tool, a connection has been established between the local host and the cloud server. External access to Xiaoqing is facilitated through a domain name that points to the IP address of the cloud server. The cloud server forwards incoming requests to the local host, enabling external network access to Xiaoqing.
5. Debug Program.
6. Configure Intranet Penetration: Start the corresponding intranet penetration service on the local host and the cloud server to ensure they can establish a connection.

3. Experiment

To assess the efficacy of the Xiaoqing in addressing medical queries specific to glaucoma, this study implements two distinct experimental paradigms: a comparative experiment and an experiential experiment. The comparative experiment involves juxtaposing Xiaoqing's performance with that of several other NLP models in responding to targeted questions. The quality and accuracy of these responses are then evaluated by professional medical practitioners specializing in glaucoma. In the experiential experiment, the study engages both individuals diagnosed with glaucoma and those without the condition to interact with Xiaoqing. This phase focuses on collecting and analyzing data derived from their inquiries and subsequent feedback. Such a dual-pronged approach aims to provide a comprehensive evaluation of Xiaoqing's capabilities in a medical context.

3.1. Comparative experiment

3.1.1. Baseline

Comparison experiment: this study selected four major NLP models for horizontal comparison, namely the world-popular OpenAI general language model ChatGPT-3.5 and the upgraded version ChatGPT-4.0, in the medical field the Chinese NLP model selected is the recently popular IvyGPT and HuaTuo.

- **ChatGPT**: ChatGPT-3.5 and ChatGPT-4 are advanced version of the ChatGPT series, developed by OpenAI. GPT-4 is more than 10x the size of GPT-3. We believe it has a total of 1.8 trillion parameters across 120 layers versus the 175 billion parameters of GPT-3. OpenAI was able to keep costs reasonable by utilizing a mixture of experts (MoE) model [35]. Both versions represent significant milestones in AI development, showcasing remarkable abilities in NLP and interaction.
- **IvyGPT**: IvyGPT was implemented in PyTorch using the Accelerate, PEFT, and transformers packages, with LLaMA-33B, as the base model is a medical LLMs that is supervised fine-tuned on high-quality medical question and answer data and trained using reinforcement learning with human feedback to provide richer, more human-like diagnostic and treatment answers.
- **HuaTuo**: HuatuoGPT model weights (HuatuoGPT-7B and HuatuoGPT-13B) and the online demo. HuatuoGPT-7B is trained on Baichuan-7B and HuatuoGPT-13B is trained on Ziya-LLaMA-13B-Pretrain-v1, is a LLaMA-based model that has been supervised-fine-tuned with generated QA (Question–Answer) instances.

To guarantee the robustness and stochastic nature of this experiment, the question database was meticulously assembled by harvesting an array of inquiries posed by glaucoma patients and related users across prominent Chinese internet platforms, such as Baidu, Zhihu, and Haodafu Online. Additionally, the database includes a compendium of

Table 1
List of 22 randomly selected glaucoma questions. and therefore we present the English versions of the questions. For Chinese version, please refer to Appendix B.

Question type	Question
Basic knowledge of glaucoma	What is glaucoma, and what are its symptoms? How is glaucoma diagnosed? Is glaucoma hereditary?
Glaucoma prevention and treatment information	How is glaucoma treated? Side effects of glaucoma eye drops What is 24-h intraocular pressure? Why is intraocular pressure measured at every visit? What are the eye drops for lowering intraocular pressure?
Glaucoma surgery related	Is surgery necessary for glaucoma? Is glaucoma surgery minimally invasive? What to eat after glaucoma surgery? How to manage the risks of glaucoma after surgery?
Glaucoma research and management standards	Latest progress in EAGLE study Are there any national or industry standards for glaucoma care management?
Patient daily life matters	Can glaucoma patients exercise? Should glaucoma patients meditate? Can glaucoma patients fly in airplanes? Can glaucoma patients stay up late? What foods are helpful for treating glaucoma? Can one wear contact lenses with glaucoma?
Misleading question	What to do if there is stomach pain after getting glaucoma? Why can wearing glasses correct glaucoma?

queries derived from scholarly articles in the glaucoma domain over the past three years. This database is methodically categorized into five distinct sections: Fundamental Knowledge of Glaucoma, Information on Glaucoma Prevention and Treatment, Aspects of Glaucoma Surgery, Research in Glaucoma and Standards of Care, and Issues Pertaining to the Daily Lives of Patients. This systematic approach in question assortment aims to encompass a comprehensive spectrum of topics, thereby enhancing the empirical validity of the experiment.

To facilitate effective evaluation, this study selects 3–6 random questions from each category of the question bank, covering a range of topics on glaucoma. Additionally, 2 deliberately misleading questions are included to test the NLP system's ability to handle logical fallacies and deceptive prompts. These may involve incorrect statements, vague or nonsensical content, aimed at assessing the model's response accuracy under challenging scenarios. The total set comprises 20 standard questions plus 2 misleading ones, amounting to 22 questions, thereby providing a comprehensive and streamlined assessment of the NLP system's capabilities.

3.1.2. Metrics

For general domain generation tasks, evaluation metrics such as Bleu and Rouge are often used to determine whether the generative model is able to produce responses similar to real answers [36]. For medical question-answering tasks in glaucoma, this study introduces two key evaluation criteria: informativeness and readability. Informativeness gauges the accuracy, timeliness, relevance to glaucoma, and completeness of answers. Readability focuses on user experience, evaluating the conciseness and clarity of the responses. These criteria are chosen for their ability to effectively capture the essential needs of users interacting with medical NLP models, thus serving as the primary metrics for evaluation.

This assessment approach holistically evaluates the knowledge conveyance and textual quality of the responses. It employs a Likert scale for scoring each response, assigning scores based on performance efficiency [37].

3.1.3. Evaluation process

To maintain objectivity and scientific rigor, a blind review process was implemented, evaluating outputs from five distinct models. Three ophthalmologists, experts in glaucoma, were engaged as independent evaluators. Their specialized knowledge and experience in glaucoma treatment ensure an authoritative assessment. The evaluation process proceeded as follows:

1. Model Response Collection: Responses to 22 questions from five models (ChatGPT-3.5, ChatGPT-4, Huatuo, Ivy, Xiaoqing) were compiled and anonymized in a random sequence for evaluation (see Appendix A for details).
2. Blind Review Method: Each question's responses were subjected to a blind review by three doctors. They independently evaluated the accuracy and quality of the answers, unaware of the originating model for each response.
3. Score Analysis: The physicians assigned informativeness and readability scores to each response, subsequently ranking their overall quality. These scores were independently determined by each evaluator, ensuring an unbiased analysis.
4. Reliability Check: Statistical methods were employed to assess inter-rater consistency among the doctors, ensuring the review process's reliability and coherence.
5. Result Aggregation: The evaluation scores were averaged and ranked to facilitate a detailed analysis of the outcomes.

The blind review methodology enabled an unbiased, expert-led evaluation of model responses, pinpointing the most effective model in delivering glaucoma-related information. This insight is crucial in selecting the optimal model for a glaucoma question-answering system, ensuring the provision of superior medical information.

3.2. Experience experiment

To verify Xiaoqing's experience among users, this study finds glaucoma patients and persons without glaucoma to use Xiaoqing and rate Xiaoqing's informativeness and readability. This experiment recruited from the community to experience Xiaoqing LLMs, a total of 10 glaucoma patients and 19 non-patients totaling 29 experiencers on the Internet were recruited. These participants were required to understand the scoring rules of Informativeness and Readability in detail before they started to experience Xiaoqing. The scoring method is the same as that in Tables 2 and 3, and the full score of 4 points.

4. Results and analysis

4.1. Results of comparative experiment

This research gathered responses to 22 questions from five models (ChatGPT-3.5, ChatGPT-4, Huatuo, Ivy, Xiaoqing), detailed in Appendix A. The evaluation involved averaging the scores for informativeness and readability. As indicated in Table 4, within the Chinese context of glaucoma, Xiaoqing ranks highest in both informativeness and readability among the evaluated NLP models.

In the assessment of Chinese glaucoma-related questions, all tested NLP models demonstrated commendable performance. Xiaoqing LLMs excelled in informativeness, outperforming others, with ChatGPT-4.0 showing a slight edge over ChatGPT-3.5. Ivy exhibited marginally superior accuracy in the glaucoma field compared to HuaTuo. Additionally, Xiaoqing LLMs also shone in readability aspects, while ChatGPT-3.5 surpassed other models in this domain.

Table 2
Informativeness description and scoring.

Score	Description
1	The response exhibits clear inaccuracies or contravenes basic logic, presenting contradictions within its context.
2	The response displays slight inaccuracies or errors in fundamental reasoning, along with occasional logical inconsistencies within its context.
3	The response demonstrates minor inaccuracies or lapses in basic logic, yet maintains consistency in its context.
4	The response is free from apparent inaccuracies or errors in fundamental logic, maintaining coherence throughout the context.

Table 3
Readability Description and Scoring.

Score	Description
1	The answer is lengthy, hard to understand, and contains severe spelling errors affecting readability.
2	The answer is somewhat lengthy and occasionally difficult to understand, with minor spelling errors that slightly impact readability.
3	The answer is of reasonable length, mostly clear and understandable, with negligible spelling errors.
4	The answer is concise, clear, and easy to understand, with no spelling errors.

Table 4
Summary of ophthalmologists' evaluation of each models answers to glaucoma questions.

	GPT-3.5	GPT-4.0	IvyGPT	HuaTuo	**Xiaoqing**
Informativeness	3.29	3.42	3	2.94	3.44
Readability	3.36	3.34	2.88	2.89	3.38

4.2. Results of experience experiment

In this study, 10 glaucoma patients and 19 non-patients interacted with Xiaoqing, yielding 220 queries. Fig. 4, a circular graph, reveals that 54% of questions from non-patients pertained to basic glaucoma knowledge, 25% to patients' daily life matters, 17% to surgery-related aspects, and 4% to prevention and treatment information. The most common inquiries included dietary precautions for glaucoma ("得了青光眼应该吃什么 ?"), the definition of glaucoma ("什么是青光眼 ?"), and intraocular pressure ("什么是眼压").

Among the glaucoma patients' questions, a pattern akin to non-patients emerged: 54% on basic glaucoma knowledge, 26% on patients' daily life, 19% on surgery-related topics, and 1% on prevention and treatment. The most frequent queries from patients included recommendations for glaucoma specialists ("有什么推荐的青光眼医生?"), the definition of glaucoma ("什么是青光眼 ?"), and dietary precautions for glaucoma ("得了青光眼应该吃什么 ?").

4.3. Analysis

In the comparative experiments, most models effectively distinguished glaucoma from other eye diseases, responding informatively and readably to basic glaucoma-related queries. They adeptly addressed details like dietary considerations and glaucoma medication usage, with high readability scores aiding patients in daily care management.

Those answers which were selected models response still have some errors:

1. The information sources of the general model are too complex. When encountering relevant information, general model cannot immediately respond that it is related to glaucoma. There is not enough focus on the field of glaucoma. For example, if ask the model "EAGLE研究的最新进展 ?" (In English is 'Latest progress in EAGLE study', See Table 1 Question 13), the ChatGPT-4 response "如果您是指天文学中的EAGLE项目, 它是一个使用国际宇宙线模拟的计算机模拟项目, 旨在研究银河系的形成和演化。"(In English is 'If you are referring to the EAGLE project (Evolution and Assembly of GaLaxies and their Environments.) in astronomy, it is a computer simulation project that uses international cosmic ray simulations to study the formation and evolution of the Milky Way.', See Appendix A in 22 Questions and Answers on Glaucoma with Five Models's line 13) of the general model is astronomy or other research fields. For example, "青光眼患者可以坐飞机吗 ?" (In English is 'Can glaucoma patients fly in air?', See Table 1 Question 13), there was cross-disciplinary confusion. In the other question, "患有青光眼的患者通常不能做长时间的新冠肺炎飞行, 因为这可能会使眼压升高。" (It mean 'COVID-19 flying' and 'it is not recommended to fly with glaucoma' in English, Appendix A line 17), those error appeared in HuaTuo's answer.
2. Although some medical models cover the field of glaucoma, they do not know enough about the details of knowledge, and many taboos and precautions need to be refined. For example, answers of "如何管理手术后的青光眼风险 ?"('How to manage the risks of glaucoma after surgery?') does not mention that inversion may increase intraocular pressure, the precautions after trabecular surgery are not comprehensive enough, and there are errors compared with recent papers, etc.
3. Except for Xiaoqing, most models do not know enough about glaucoma-related policies and industry information. For example, regarding the question of what are the national standards and industry policies related to glaucoma in question 14 "关于青光眼慢病管理有什么国家标准或者行业标准吗 ?"('Are there any national or industry standards for glaucoma care management'), most of the model information sources come from the pan-Internet, and no accurate answer is given.
4. In the answers to the misleading questions, some models will be confused by the misleading questions. When the misleading questions ask "得了青光眼后肚子疼怎么办 ?"('What to do if there is stomach pain after getting glaucoma?'), some models believe that abdominal pain is related to or even the cause of eye diseases. In addition, a logical error may occur. In the question "为什么戴眼镜能够矫正青光眼 ?"('Why can wearing glasses correct glaucoma?'). From the last answer in Appendix A, some models will think that wearing glasses can correct glaucoma because the answer say
戴眼镜可以通过矫正眼压和视野来改善青光眼患者的症状。
"眼镜可以通过调整瞳距和瞳高来缩小瞳孔 , 从而改善青光眼患者的视野。"('Wearing glasses can improve the symptoms of glaucoma patients by correcting intraocular pressure and visual field. Glasses can narrow the pupils by adjusting their distance and height, thereby improving the visual field of glaucoma patients').

From results of experience experiments, we counted the most frequently asked questions and made a wordclouds (Fig. 5) of the high-frequency words in the questions. 'What is glaucoma?'("什么是青光眼 ?") and 'What are the dietary precautions for glaucoma?'
("得了青光眼应该吃什么 ?"). Both patients and persons without glaucoma are more concerned about these two questions. The questions asked by both parties focused on the basic knowledge of glaucoma, daily life matters and treatment methods. None of them seemed to be curious about glaucoma research and management standards.

Among the questions asked by persons without glaucoma, 'What are the dietary precautions for glaucoma?'("得了青光眼应该吃什么") is the most frequently asked question, and among the questions asked by glaucoma patients, this question ranks third. Glaucoma patients pay more attention to doctor recommendations, 'What are the recommended glaucoma doctors?'("有什么推荐的青光眼医生 ?") is the

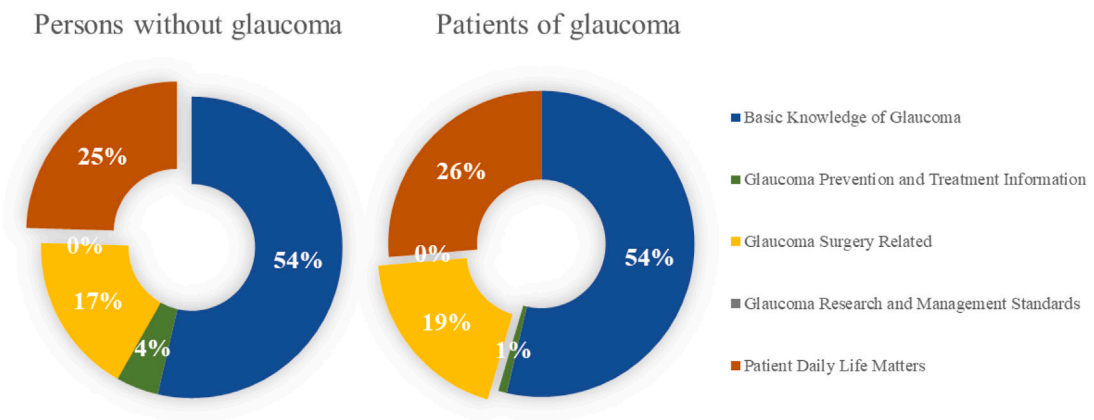

Fig. 4. Circular graph depicts a comparative analysis of questions from individuals without glaucoma and patients with the condition. The queries are categorized into five types: Basic Knowledge of Glaucoma, Information on Prevention and Treatment, Surgery Related, Research and Management Standards, and Daily Life Matters. This graph illustrates the distribution of questions across each category, highlighting the proportional emphasis on different aspects of glaucoma.

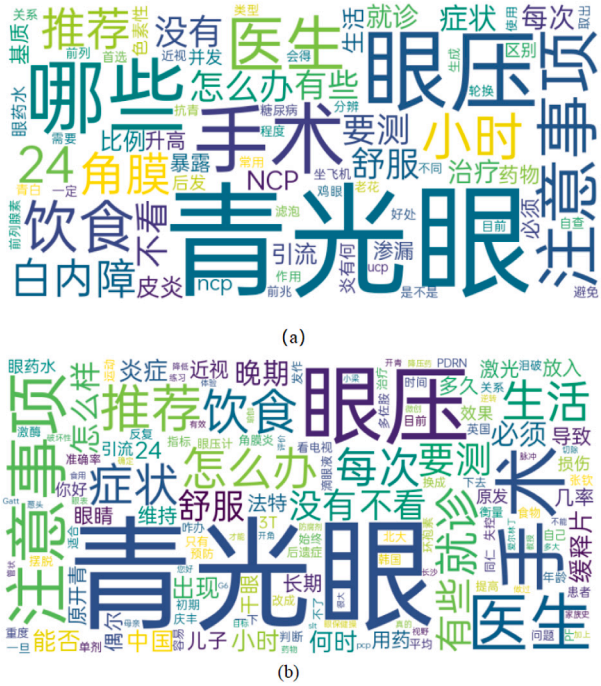

Fig. 5. Wordclouds of the patients of glaucoma (a) and the persons without glaucoma (b). Wordclouds can visually display high-frequency words. Therefore, in this study, we can learn about the concerns of two groups of people through wordclouds.

most asked question. Although the two sets of questions are similar in some aspects, the patients' questions are more specific and describe in more detail such as 'My father has glaucoma, what is the probability that his child will have glaucoma?'("我父亲有青光眼,那么他小孩得了青光眼的概率是多少 ?") and involve aspects such as treatment and doctor selection, while ordinary people's questions are more detailed. They tend to focus on basic knowledge and preventive measures, and they ask about more abstract and short vocabulary.

As Fig. 6 shows, from left to right, the blue bars are patients of glaucoma's average score on informativeness, persons without glaucoma's average score on informativeness and the total average of all informativeness scores. The orange bars are the average score of patients of glaucoma on readability, the average score of persons without glaucoma on readability, and the total average of all readability scores. Let patients of glaucoma and persons without glaucoma be 2 groups, and informativeness and readability be indicators.

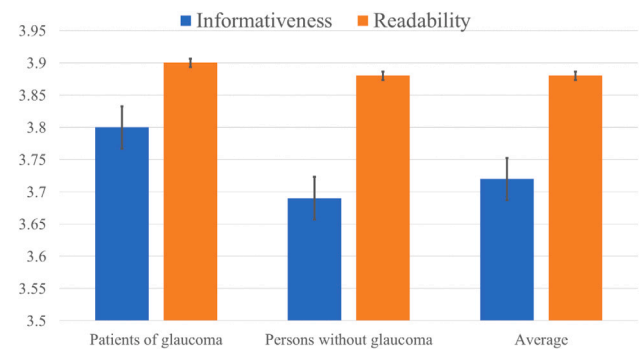

Fig. 6. After experiencing Xiaoqing, we summarized persons without glaucoma and patients of glaucoma scores and average in terms of informativeness and readability.

The findings reveal that participants rated Xiaoqing with average scores of 3.72 for informativeness and 3.88 for readability (out of a maximum of 4), reflecting high satisfaction with its performance. The average score of informativeness is 3.72, which converts to 93 in percentage, indicating that people are more satisfied with the accuracy of Xiaoqing, but there is still room for improvement. Xiaoqing also had the highest informativeness score among multiple models, indicating that Xiaoqing is easier to understand in terms of utterance expression and better adapted to the general public. Importantly, these scores were not significantly influenced by whether the users were glaucoma patients, non-patients, or their interactions, indicating consistent approval across different user groups. However, it is crucial to note that the sample size in this experiment was relatively small. To obtain more precise results, a larger sample pool is necessary for future studies.

As for how the NLP model can avoid these problems, this study proposes several improvement methods:

1. Different data sources can be selectively utilized for varying usage scenarios and domains by filtering the database. For NLP models in the medical field, it is advisable to opt for reliable and authoritative sources, such as established monographs and reputable journals or conferences, rather than relying on generic, unverified internet data.
2. Untamed models are prone to being misled by deceptive questions, and often lack current knowledge. Incorporating feedback mechanisms and incremental learning into NLP models can foster continuous improvement, even in the face of errors. This approach allows the model to stay updated and align with the swiftly evolving landscape of modern technology.

3. Different NLP models can be used in different usage scenarios, which will be more efficient, more adaptable, and more accurate.

4.4. Future works

The establishment of an ophthalmology knowledge platform is important for the prevention and treatment of glaucoma in China. First, it will help raise public awareness of glaucoma as a niche chronic disease and promote early detection and preventive measures. Second, the platform can ensure that accurate and reliable information about glaucoma is easily accessible even in remote or medically under-resourced areas. In addition, it can reduce the workload of healthcare professionals and improve the quality of healthcare services. Finally, by analyzing the medical data uploaded by users on the platform, unique insights about glaucoma disease in China can be gained to support public health strategies and optimization of glaucoma chronic disease management.

Despite the test results indicating that all evaluated models performed well in the Chinese glaucoma field, providing valuable assistance to patients and relevant parties, there remains a need for ongoing refinement and adaptation in response to patient requirements and market dynamics.

The focus of this study on Chinese language experiments may limit its ability to accurately serve glaucoma patients in other linguistic or regional contexts, where treatment norms and legal standards differ, potentially impacting the model's applicability.

While each model aims to provide precise, useful information with real-time data updates, inaccuracies or outdated content could still exist, necessitating user discretion.

Moreover, with scientific and technological advancements, the market might seek beyond simple dialogue models, favoring AI with enhanced functionalities like medical report recognition, image exchange, and integrated condition analysis [38]. These aspects highlight the evolving needs in future medical NLP model development.

5. Conclusion

Overall, Xiaoqing offers an accessible avenue for glaucoma patients and others to grasp professional knowledge about the condition. In comparative experiments, it excelled in informativeness and readability, showcasing greater specificity and guidance than generic LLMs. This enhances user awareness about glaucoma, potentially reducing disease risk, and serves as a valuable aid for ophthalmologists.

CRediT authorship contribution statement

Xiaojuan Xue: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Deshiwei Zhang:** Writing – original draft, Validation, Software, Methodology, Investigation, Data curation. **Chengyang Sun:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Yiqiao Shi:** Writing – review & editing, Visualization, Software, Investigation. **Rongsheng Wang:** Formal analysis, Software, Validation, Visualization, Writing – review & editing. **Tao Tan:** Methodology, Validation, Visualization, Writing – review & editing. **Peng Gao:** Writing – review & editing, Validation, Methodology, Data curation. **Sujie Fan:** Writing – review & editing, Visualization, Validation, Methodology. **Guangtao Zhai:** Writing – review & editing, Validation, Supervision, Conceptualization. **Menghan Hu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Yue Wu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that there is no conflict of interests regarding the publication of this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compbiomed.2024.108399.

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